

Write all probabilities as decimals and round to the nearest three decimal places when needed.

1) You pick 1 marble from a bag that contains 8 marbles (2 blue, 3 red, and 3 yellow). Find the following probabilities.

- a) $P(\text{Blue})$ b) $P(\text{Yellow})$ c) $P(\text{Green})$
- d) $P(\text{Blue or Yellow})$ e) $P(\text{not Blue})$ f) $P(\text{Red, Blue, or Yellow})$

2) A card is selected at random from a deck of 52 cards. Each deck has 4 different suits: Clubs (black), Spades (black), Diamonds (red), and Hearts (red). Each suit contains the following cards: Ace, 2, 3, 4, 5, 6, 7, 8, 9, 10, Jack, Queen, and a King. Find the following probabilities.

- a) $P(\text{Diamond})$ b) $P(\text{black})$ c) $P(\text{not black})$
- d) $P(\text{not Heart})$ e) $P(\text{Ace})$ f) $P(\text{number card})$
- g) $P(\text{even number})$ h) $P(8 \text{ or } 9)$ i) $P(\text{not the Queen of Hearts})$

3) Below are the probabilities of pulling out a particular color from an M&M bag. What is the probability of pulling a blue M&M? Describe the likelihood of this event.

Brown	Red	Yellow	Green	Orange	Blue
0.30	0.20	0.20	0.10	0.10	?

4) You roll two 6-sided dice. Find the probabilities of rolling the following sums. Note: Not all probabilities are the same (the probability of rolling 7 is higher than the probability of rolling 2). There are 36 possible outcomes.

	1	2	3	4	5	6
1	1,1	1,2	1,3	1,4	1,5	1,6
2	2,1	2,2	2,3	2,4	2,5	2,6
3	3,1	3,2	3,3	3,4	3,5	3,6
4	4,1	4,2	4,3	4,4	4,5	4,6
5	5,1	5,2	5,3	5,4	5,5	5,6
6	6,1	6,2	6,3	6,4	6,5	6,6

a) $P(\text{sum is } 2)$

b) $P(\text{sum is } 7)$

c) $P(\text{sum is } 2 \text{ or } 3)$

d) $P(\text{sum is } 6 \text{ or } 8)$

e) $P(\text{sum is even})$

f) $P(\text{sum is not } 3)$

g) $P(\text{sum is } 5 \text{ or greater})$

h) $P(\text{sum is not equal to } 3 \text{ or } 7)$

i) $P(\text{sum is } 10 \text{ or less})$

5) A survey of 324 people asked what their favorite food was. The results are shown below.

	Pizza	Burgers	Fried Chicken	Other	Total
Less than 18	60	23	5	34	
18 and older	45	33	20	104	
Total					

If we randomly select a person from this sample,

a) What is the probability that person liked fried chicken?

b) What is the probability that person is less than 18 years old and likes burgers?

c) What is the probability that person likes pizza or burgers?

d) What is the probability that person is less than 18 or 18 and older?

6) Tony is a great basketball player and claims he is an 80% free throw shooter. During the most recent game, he shoots 9 free throws and he only makes 5. Is he really an 80% free throw shooter? To answer this question, run a simulation to see how likely it is to make 5 out of 9 shots if he really is an 80% free throw shooter. Use the four-step process and perform at least 20 trials of your simulation.